

Appendix for 'SCI-Mamba: Unsupervised Learning based Low-Light Image Enhancement for Non-Cooperative Spacecraft'

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This appendix provides supplementary materials supporting the main text and is organized as follows. Section I details the construction of *Space Dark-1.0*, including synthetic spacecraft images generated with PASim, darkroom hardware-in-the-loop captures of a scaled spacecraft model, and real on-orbit images extracted from publicly released ADRAS-J mission videos. Section II presents a leave-one-out ablation study of the five loss terms in the multi-objective loss function. Section III reports qualitative comparisons on representative images from each category of *Space Dark-1.0*, together with a generalization test on the SPARK 2021 dataset. Section IV evaluates the inference efficiency and resource consumption of SCI-Mamba on an actual edge-computing platform. Section V presents supplementary experiments on spacecraft object detection, assessing the effects of different enhancement methods on downstream detection performance. Together, these materials corroborate the dataset construction, model design, and practical applicability of the proposed method.

I. DATASET PREPARATION

This section describes the preparation of *Space Dark-1.0*, which comprises 9,400 images from three complementary sources. Subsection I-A introduces synthetic spacecraft images generated using PASim. Subsection I-B describes images of a scaled spacecraft model captured in ground-based darkroom experiments, and Subsection I-C presents real on-orbit images extracted from publicly released ADRAS-J mission videos. These sources combine simulated orbital imaging conditions, real camera responses under low illumination, and observations from the actual space environment. The following subsections detail the generation, acquisition, and selection procedures for the respective subsets.

A. Physically Constrained Synthetic Satellite Images Generated Using PASim

This subset comprises 4,700 low-light spacecraft images generated using PASim, a spacecraft simulation platform developed in-house. Built on Unreal Engine, PASim incorporates orbital and attitude dynamics models to simulate the target's orbital motion, attitude evolution, and relative motion. For dataset generation, the target's orbital semi-major axis, inclination, and mean anomaly were set to 6,886 km, 45°,

and 90°, respectively. Once the orbital and attitude states were determined, low-light images were generated using Unreal Engine's built-in lighting and rendering capabilities, accounting for space illumination conditions and the reflective properties of spacecraft surface materials.

B. Images of a Scaled Spacecraft Model Captured in Ground-Based Darkroom Experiments

This subset comprises 2,600 images of a scaled spacecraft model acquired using a ground-based darkroom experimental platform to characterize the response of a real imaging system under low-light conditions. The experimental setup is shown in Figure 1. A 1:35 scale model of the BeiDou-1 satellite was used, and images were captured using an Intel RealSense D435i camera. During the experiments, the light source was positioned above the satellite model, while the darkroom environment reduced interference from background stray light, enabling image acquisition under different low-light conditions. The acquired images were screened to form the darkroom experimental subset. Compared with synthetic images, this subset retains characteristics of real camera imaging, including sensor noise, limited dynamic range, and brightness variations, thereby capturing practical imaging effects that are difficult to fully reproduce through simulation and rendering.

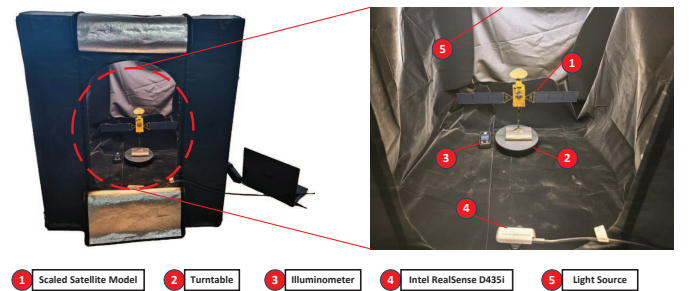


Fig. 1. Darkroom

C. Real On-Orbit Images Extracted from Publicly Released ADRAS-J Mission Videos

This subset comprises 2,100 real on-orbit images extracted from publicly released videos of the H-IIA launch vehicle upper stage captured during Astroscale's ADRAS-J mission. Individual frames were first extracted from the mission videos and then manually screened to remove duplicate frames,

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visibly blurred frames, and images containing limited target information. Representative samples spanning different target attitudes, observation distances, background conditions, and illumination states were retained. This subset incorporates imaging characteristics of the actual space environment, including complex illumination, deep-space backgrounds, and reflections from the target surface, supporting the evaluation of enhancement methods in real on-orbit scenarios. Derived directly from on-orbit observations, these samples complement the synthetic and ground-based darkroom subsets.

II. ABLATION STUDY ON LOSS FUNCTIONS

To evaluate the roles of the five loss terms in the joint optimization of SCI-Mamba, we conduct a leave-one-out ablation study. Under the same experimental conditions and using the same dataset, we set the weight of one loss term to zero at a time, while keeping the weights of the remaining terms and all other training settings unchanged, and retrain the model for each configuration. The resulting model weights are then used for inference on the same test images. All configurations share the same network architecture and inference procedure and exhibit the same inference efficiency. Therefore, this experiment focuses exclusively on how different combinations of loss terms affect image enhancement.

Figure 2 presents the ablation results for synthetic spacecraft images, ground-based darkroom images, and real on-orbit images. The columns labeled Loss1 through Loss5 show the outputs of models retrained with the corresponding loss weight set to zero, whereas SCI-Mamba denotes the model trained using all five loss terms. This experiment evaluates the contribution of each loss term to the final enhancement result in the presence of the other four constraints. Removing Loss1 introduces pronounced red or orange-red color shifts across all three image categories, particularly in the solar panels of the synthetic spacecraft, the central structure of the darkroom target, and the reflective surfaces of the on-orbit target. Removing Loss3 also produces color distortions: the solar panels in the synthetic image exhibit yellow-green mottling, while uneven yellow and purple patches appear on the on-orbit target. These observations indicate that, even with the remaining loss terms, the model can introduce substantial distortions while increasing brightness. The constraints imposed by Loss1 and Loss3 therefore cannot be fully replaced by the remaining terms. Removing Loss5 produces outputs that more closely resemble the low-light inputs across all three image categories, with substantially reduced target brightening. Improvements in the visibility of the central structure of the darkroom target and the surface of the on-orbit target are consequently limited. This finding indicates that effective enhancement requires both the suppression of color and appearance distortions and a sufficient degree of brightening. Without the corresponding constraint, the model may produce conservative outputs that avoid severe color distortions but provide insufficient low-light enhancement. Removing Loss2 or Loss4 primarily affects local brightness and color relationships. For example, the solar panels in the synthetic image exhibit different color tones from those produced by the complete model, while the reflective

regions of the on-orbit target also show changes in brightness and color. These results highlight that loss terms contribute not only to preventing conspicuous distortions but also to regulating local enhancement characteristics in conjunction with the other terms. Because the remaining four losses continue to participate in training, each ablated model is still subject to multiple constraints. Accordingly, the necessity of Loss2 and Loss4 cannot be assessed solely by whether their removal causes severe color shifts or a substantial reduction in brightness, nor does similarity in overall appearance establish that their contributions are fully replaced by the remaining terms. Taken together, these results indicate that the five loss terms form a joint optimization framework with complementary and mutually constraining roles, rather than acting as independent, interchangeable objectives. Removing any term changes the combination of training constraints and shifts the balance among the remaining objectives. This shift may manifest as pronounced color distortion, insufficient enhancement, or changes in local brightness and color relationships. The contributions of the individual terms therefore should not be ranked solely by the visual magnitude of the changes observed in the ablation results. Retaining all five terms helps balance target brightening, color rendition, and local appearance, providing the rationale for using the complete loss formulation.

III. INFERENCE EXPERIMENT

This section provides the qualitative comparison results omitted from the main manuscript due to space constraints. The main manuscript focuses on the quantitative comparison of inference efficiency and resource consumption; the visual comparisons of enhancement outputs are presented here. Subsection III-A tests a synthetic virtual orbital picture from *Space Dark-1.0*, Subsection III-B a darkroom hardware-in-the-loop capture of a scaled satellite model, and Subsection III-C an authentic on-orbit spacecraft picture. Subsection III-D further tests the SPARK 2021 dataset [1], which contains previously unseen satellite types not used in training.

The comparisons focus on changes in target brightness, the visibility of dark structures, background noise, and the appearance of bright regions. The following subsections discuss both the improvements and limitations observed in the displayed examples.

A. Test on Virtual Pictures

We first evaluate the enhancement results on a synthetic image from *Space Dark-1.0*. Figure 3 presents the outputs of the compared methods under simulated orbital illumination conditions.

In Figure 3, the main target's body and solar panels are already recognizable in the input, whereas the smaller target near the bottom of the image remains dim. SCI++ [2] and Zero-DCE++ [3] increase target brightness, while Uformer produces a comparatively limited change. LLFormer also brightens the target, accompanied by a visible increase in background intensity. RUAS [4], LLFlow [5], ECMamba [6], WalMaFa [7], and UHDformer [8] produce stronger brightening in several target regions, with particularly pronounced whitening in the ECMamba and UHDformer outputs.



Fig. 2. Ablation Study on Loss Components

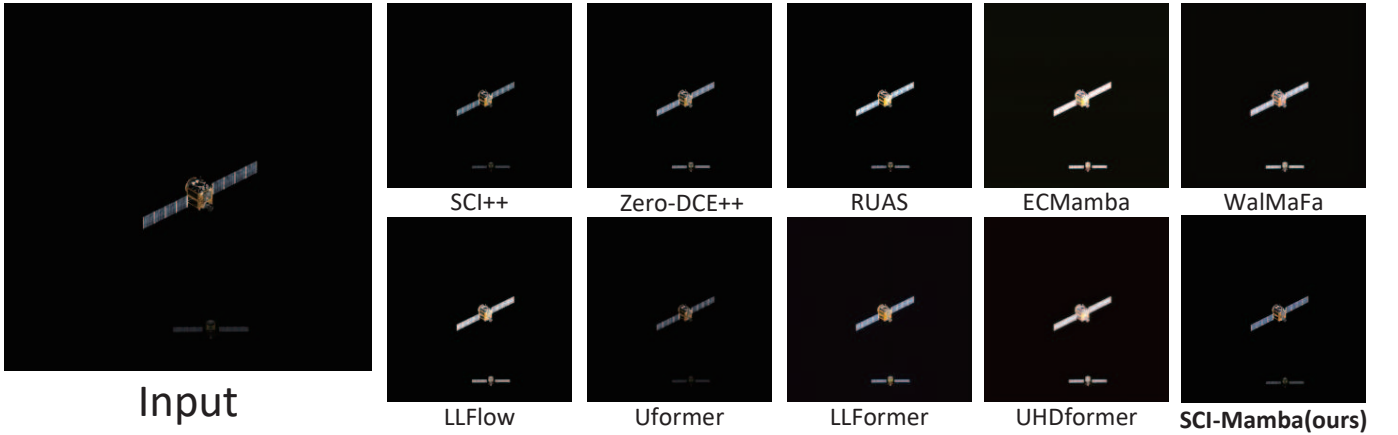


Fig. 3. Visual Comparison on Synthetic Virtual Orbital Picture

SCI-Mamba increases the brightness of the main target while keeping the background dark. The principal body contours and solar-panel divisions remain discernible after enhancement. However, some reflective features become more pronounced, and the smaller target near the bottom of the image remains relatively dim. This example shows that SCI-Mamba improves the visibility of the main target without conspicuous background brightening, although the improvement is less evident for the darker, smaller target.

B. Test on Real Pictures

Figure 4 compares the enhancement results for a low-light image of a scaled satellite model captured in the ground-based darkroom. The target is weakly illuminated, and visible noise is present in the input. This example illustrates the trade-off between improving the visibility of dark structures and limiting the amplification of background noise.

SCI++, Uformer, and LLFormer produce relatively dark

outputs, in which the solar panels remain difficult to distinguish. RUAS increases target brightness but also makes background noise more apparent. Zero-DCE++, ECMamba, and WalMaFa reveal more of the solar panels and surrounding structures, accompanied by stronger visible background noise and artifacts. LLFlow and UHDformer produce particularly pronounced background brightening and colored artifacts, with predominantly magenta and reddish appearances, respectively. SCI-Mamba makes the central body and some supporting structures more visible, while background noise is less conspicuous than in several of the more strongly enhanced outputs. However, the solar panels remain dark, and some reflective edges become disproportionately bright. In addition, some reddish artifacts are noticeable along the lower edge of the spacecraft body. The result therefore shows a benefit in limiting background amplification, together with limited recovery of the darkest target structures.

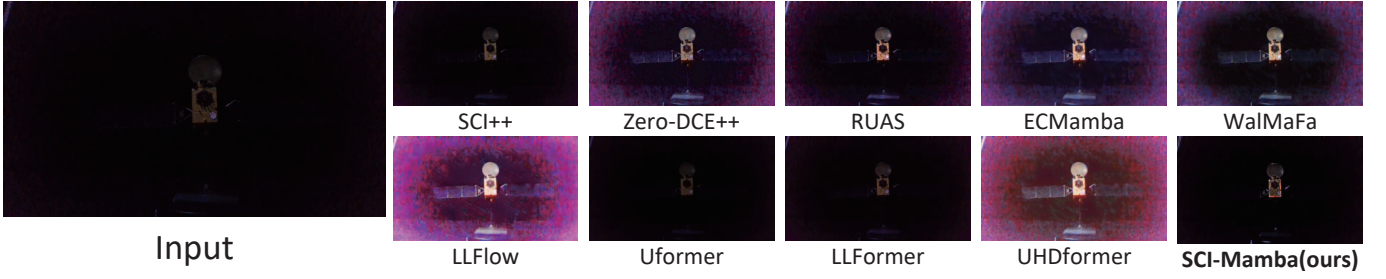


Fig. 4. Visual Comparison on Darkroom Captured Satellite Picture

C. Test on On-Orbit Spacecraft Pictures

Figure 5 presents a qualitative comparison for an on-orbit image of a launch vehicle upper stage. The input contains reflective target surfaces, darker structural regions, and visible background noise. This example allows the effects of enhancement on bright surface regions and the surrounding background to be examined together.

The input already shows a recognizable target outline and several surface structures. SCI++ increases target brightness while retaining discernible surface detail, whereas Uformer produces a more limited brightness change. RUAS, ECMamba, and UHDformer produce pronounced whitening in reflective regions, making some local surface details less distinguishable. Zero-DCE++, WalMaFa, and LLFormer brighten the target but also make background noise more conspicuous. LLFlow exhibits prominent horizontal streaks and block-like artifacts. SCI-Mamba increases target brightness without the marked background brightening seen in several competing outputs, and the main target outline remains recognizable. However, some reflective surfaces and edges become excessively bright, reducing the visibility of local detail. The central band also appears brighter and more yellowish than in the input. These observations indicate that the result combines limited background amplification with a remaining trade-off between target brightening and the preservation of detail in bright regions.

D. Generalization Test on the SPARK 2021 Dataset

To evaluate the generalization capability of SCI-Mamba on unseen space imagery, we further conduct experiments on the SPARK 2021 dataset [1]. The images in the SPARK 2021 dataset are generated in a realistic space simulation environment with highly diverse sensing conditions, including extreme and challenging scenarios across different orbits, background noise, low signal-to-noise ratios (SNRs), and the high image contrast characteristic of real space imagery. The corresponding visual comparison is presented in Figure 6.

In Figure 6, the spacecraft is surrounded by a dark space background, with a bright Earth region in the lower-right portion of the image. The body and solar panel are recognizable in the input, although their brightness differs substantially. SCI++ brightens the spacecraft body while keeping the background relatively dark, whereas Uformer produces a comparatively limited brightness change. Zero-DCE++, ECMamba, WalMaFa, LLFlow, LLFormer, and UHDformer increase the brightness of the solar panel to varying degrees, but also make

background noise more conspicuous. LLFlow and LLFormer exhibit particularly noticeable colored background fluctuations. RUAS strongly brightens the target and the Earth region, with pronounced whitening in parts of the spacecraft and cloud cover.

SCI-Mamba primarily increases the brightness of the spacecraft body and its brighter appendages, while the large solar panel changes less and remains relatively dark. The background does not undergo the broad intensity increase seen in several competing outputs, although visible noise remains. The Earth region is also affected by enhancement, but its overall brightening is less pronounced than in the RUAS output. This example provides qualitative evidence of the method’s enhancement behavior on SPARK imagery, while also showing uneven enhancement across target regions. Broader conclusions about generalization require evaluation over more samples and imaging conditions.

Across the four displayed examples, SCI-Mamba increases the brightness of selected target regions while generally limiting the broad background brightening observed in several competing outputs. Its effect on structural visibility, however, varies across scenes and target regions. Very dark structures remain insufficiently visible in the darkroom example, while some reflective regions become excessively bright in the on-orbit example. The SPARK result provides an additional qualitative example of this enhancement behavior. These comparisons illustrate both the benefits and limitations of the method; they do not establish uniformly improved detail preservation or generalization across all space imaging conditions.

IV. INFERENCE EVALUATION ON AN EDGE-COMPUTING PLATFORM

To further assess the feasibility of deploying SCI-Mamba on edge-computing devices, we extended the V100-based evaluation reported in the main manuscript by deploying the algorithm on an NVIDIA Jetson Orin NX 16GB and conducting inference using pretrained weights. The evaluation included 1,400 images from the Space Dark-1.0 test set and 100 images from the SPARK dataset, totaling 1,500 images. SCI-Mamba completed inference on all test images at an average speed of 5.25 FPS, demonstrating its ability to perform image enhancement on the evaluated edge device. We also deployed SCI++ on the same Jetson Orin NX 16GB and evaluated it using the same test data, obtaining an average inference speed of 71.39 FPS. Compared with the corresponding V100 results reported in Table II of the main manuscript, both

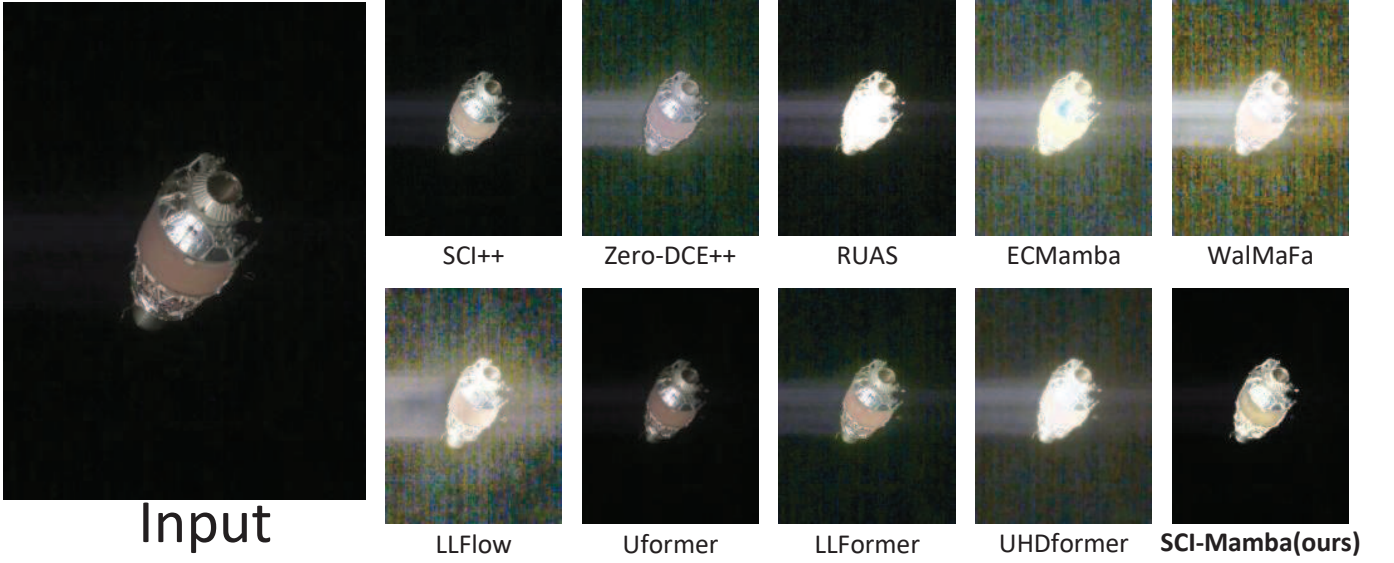


Fig. 5. Visual Comparison on Medium-bright Real On-orbit Spacecraft Picture

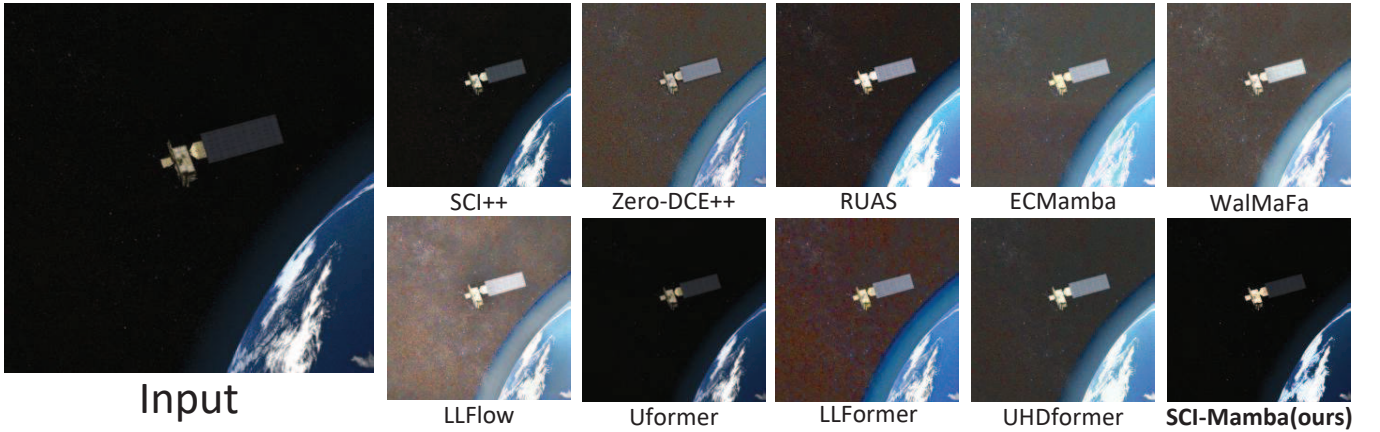


Fig. 6. Visual Comparison of Different Low-Light Image Enhancement Methods on the SPARK 2021 Dataset

methods exhibited lower frame rates on the Jetson Orin NX, consistent with the differences in hardware resources between the two platforms. The V100 provides 5,120 CUDA cores and a memory bandwidth of 900 GB/s, whereas the Jetson Orin NX 16GB provides 1,024 CUDA cores and a memory bandwidth of approximately 102 GB/s. These specifications indicate that the edge platform has more limited resources for parallel computation and memory access. The reduction in inference frame rate following migration from the V100 to the Jetson Orin NX is therefore consistent with the expected performance impact of these resource constraints. However, differences in GPU architecture and operating conditions mean that ratios of core counts and memory bandwidth cannot be directly translated into ratios of inference frame rates; the magnitude of the reduction also depends on operator execution efficiency and runtime configuration. By completing inference on the entire evaluation set, this experiment provides additional empirical evidence of SCI-Mambas processing capability on an edge platform and further supports its deployment feasibility under the evaluated hardware and configuration.

V. SUPPLEMENTARY EXPERIMENTS ON SPACECRAFT OBJECT DETECTION

A. Experimental Objective and Settings

To evaluate the effects of low-light image enhancement on spacecraft object detection, we conduct comparative experiments using YOLOv8 [9]. Detection results obtained from images processed by different enhancement methods are compared to assess their effects on spacecraft detection and localization performance.

A subset of 300 images is randomly selected from the held-out test set of Space Dark-1.0, which was not used in training the enhancement model, and divided into training, validation, and test sets containing 210, 60, and 30 images, respectively. The YOLOv8n detector is initialized with pretrained weights and trained for 100 epochs, with the input image size parameter set to 640 and a training batch size of 10.

For the comparative evaluation, SCI-Mamba, SCI++, Uformer, LLFormer, RUAS, Zero-DCE++, UHDformer, ECMamba, WalMaFa, and LLFlow are applied to the same test images. Each enhanced image is matched to its corre-

sponding annotations by filename. All methods are evaluated using the same trained detector weights, object categories, and bounding-box annotations, with the input image size parameter set to 640 and a batch size of 2. The validation set was used for model selection during training, and all comparative results reported below are obtained on the held-out test set of 30 images.

B. Detection Metrics and Auxiliary PR-HV Analysis

Detection performance is evaluated using four standard metrics: Box Precision, Box Recall, mAP50 [10], and mAP50–95 [11]. Precision measures the proportion of detections that are correct, whereas Recall measures the proportion of ground-truth objects successfully detected. mAP50 denotes the mean average precision at an intersection over union (IoU) threshold of 0.50. mAP50–95 averages the results over IoU thresholds ranging from 0.50 to 0.95 in increments of 0.05, thereby evaluating detection performance under different localization requirements.

Drawing on the hypervolume indicator [12], we further employ a point-wise precision–recall hypervolume score (PR-HV) to quantify the joint performance of Precision and Recall. With the origin as the reference point, the score is expressed as

$$\text{PR-HV}_i = P_i R_i \times 100\%, \quad (1)$$

where P_i and R_i denote the Box Precision and Box Recall associated with the i th enhancement method, respectively, both expressed as values in $[0, 1]$ for calculation. PR-HV summarizes their joint performance through their product and is reported alongside the standard detection metrics as an auxiliary measure for multi-objective analysis.

C. Quantitative Comparison

Table I presents the spacecraft object detection results obtained using different enhancement methods. P, R, mAP50, mAP50–95, and PR-HV are all reported as percentages.

SCI-Mamba achieves a Recall of 96.65% and an mAP50 of 96.52%, both the highest among the compared methods, with a Precision of 90.62%. Compared with SCI++, it improves Precision, Recall, and mAP50 by approximately 1.32, 1.23, and 0.42 percentage points, respectively. These results show that images enhanced by SCI-Mamba yield a higher object detection rate and mAP50 when evaluated using the same detector and image set.

For mAP50–95, SCI-Mamba achieves 76.40%, ranking third behind SCI++ at 80.52% and Uformer at 78.08%. SCI-Mamba obtains the highest mAP at an IoU threshold of 0.50, whereas SCI++ and Uformer achieve higher scores when performance is averaged across multiple IoU thresholds that include stricter localization requirements.

The comparison of Precision and Recall further illustrates the differences among the methods. UHDformer achieves the highest Precision of 93.54%, but its Recall is 70.00%. In comparison, SCI-Mamba achieves a Precision of 90.62% and a Recall of 96.65%. Combining these two metrics, SCI-Mamba obtains a PR-HV of 87.59%, exceeding SCI++ at 85.21% and Uformer at 84.49% and ranking first among the compared methods. This result reflects its ability to achieve high recall while maintaining high precision.

These experiments demonstrate that images enhanced by SCI-Mamba can be effectively used for spacecraft object detection. Its leading Recall, mAP50, and PR-HV in the present comparison provide experimental evidence supporting the use of SCI-Mamba as an image enhancement front end for spacecraft object detection under low-light conditions.

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TABLE I
COMPARISON OF SPACECRAFT OBJECT DETECTION PERFORMANCE USING DIFFERENT LOW-LIGHT ENHANCEMENT METHODS

Methods	P (%)	R (%)	mAP50 (%)	mAP50-95 (%)	PR-HV (%)
SCI++	89.30	95.42	96.10	80.52	85.21
Zero-DCE++	89.18	75.71	84.63	67.03	67.52
RUAS	88.77	82.86	91.24	70.47	73.55
ECMamba	82.29	70.00	74.22	55.77	57.60
WalMaFa	81.81	65.71	69.37	54.60	53.76
LLFlow	85.97	61.43	69.04	52.02	52.81
Uformer	92.41	91.43	95.59	78.08	84.49
LLFormer	92.10	88.57	91.84	74.38	81.58
UHDformer	93.54	70.00	81.47	60.69	65.48
SCI-Mamba (ours)	90.62	96.65	96.52	76.40	87.59